CMSC 4200 - Artificial Intelligence
Spring 2026, Group Assignment 4

Due – Thursday April 9, 2026, at 11:59pm

Group #	1
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Use this document to provide your answers. This assignment will be graded out of 20 points. Submit your answer file to the right D2L Dropbox named "Assignment 4 Dropbox" by the due date and time.

Question 1. (8 points) Show the correctness/incorrectness of the following:

a) $(X \wedge \neg Y) \models (X \Leftrightarrow Y)$
This is incorrect because $(X \Leftrightarrow Y)$ is not true in all worlds that $(X \wedge \neg Y)$ is true.

b) $(X \Leftrightarrow Y) \models (\neg X \vee Y)$
This is correct because $(\neg X \vee Y)$ will be true in every world that $(X \Leftrightarrow Y)$ is true.

c) $(A \vee B) \wedge (X \vee \neg Y \vee Z) \models (X \vee Z \vee \neg Y)$
This is correct because $(X \vee Z \vee \neg Y)$ will be true in every world that $(A \vee B) \wedge (X \vee \neg Y \vee Z)$ is true.

d) $(X \Leftrightarrow Y) \wedge (\neg X \vee Y)$ is satisfiable.
This is correct because there is at least one model in which this sentence is true.

Question 2. (6 points) Convert the following sentences to CNF form. Show your steps.

a) $(\neg A \wedge \neg B) \Rightarrow C$
b) $A \Leftrightarrow (\neg B \wedge \neg C)$

A) Steps	Converted
$(\neg A \wedge \neg B) \Rightarrow C$ DeMorgan's: $\neg(A \vee B)$ $\neg(A \vee B) \Longrightarrow C$ Implication Elimination: $\neg(A \vee B) \Longrightarrow C$ $\neg\neg(A \vee B) \vee C$ Double Negative: $(A \vee B) \vee C$	$(A \vee B) \vee C$

B) Steps	
$A \Leftrightarrow (\neg B \wedge \neg C)$ DeMorgan's: $\neg(B \vee C)$ $A \Leftrightarrow \neg(B \vee C)$ Biconditional: $\big(A \Rightarrow \neg(B \vee C)\big) \wedge ((B \vee C) \Rightarrow A)$ Implication Elimination: $\big(\neg A \vee \neg(B \vee C)\big) \wedge ((B \vee C) \vee A)$ DeMorgan's: $\big(\neg A \vee (\neg B \wedge \neg C)\big) \wedge (B \vee C \vee A)$ Distribute: $\big(\neg A \vee \neg B) \wedge (\neg A \vee \neg C)\big) \wedge (B \vee C \vee A)$	$\big(\neg A \vee \neg B) \wedge (\neg A \vee \neg C)\big) \wedge (B \vee C \vee A)$

Question 3. (6 points) A knowledge base *KB* has 4 sentences:

KB = {*X*, ¬*X* ∨ *Y*, ¬*X* ∨ *Z*, ¬*Y* ∨ ¬*Z* ∨ *Q*}

Does *KB* entail *Q*? Show your steps.

First, apply logical equivalences:

1. *X (No Change)*
2. *¬X ∨ Y ≡ X ⇒ Y (Implication Elimination)*
3. *¬X ∨ Z ≡ X ⇒ Z (Implication Elimination)*
4.
   a. *¬Y ∨ ¬Z ∨ Q ≡ (¬Y ∨ ¬Z) ∨ Q (Associativity)*
   b. *(¬Y ∨ ¬Z) ∨ Q ≡ ¬(Y ∧ Z) ∨ Q (De Morgan)*
   c. *¬(Y ∧ Z) ∨ Q ≡ (Y ∧ Z) ⇒ Q (Implication Elimination)*

So, the KB can be written as:

KB = {X, *X ⇒ Y, X ⇒ Z, (Y ∧ Z) ⇒ Q*}

1. Since X is given in the KB, X is true.
2. Since X is true, Y is true (from *X ⇒ Y*).
3. Since X is true, Z is true (from *X ⇒ Z*).
4. Since Y and Z are true, Q is true (from *(Y ∧ Z) ⇒ Q*).

Therefore, yes, KB does entail Q